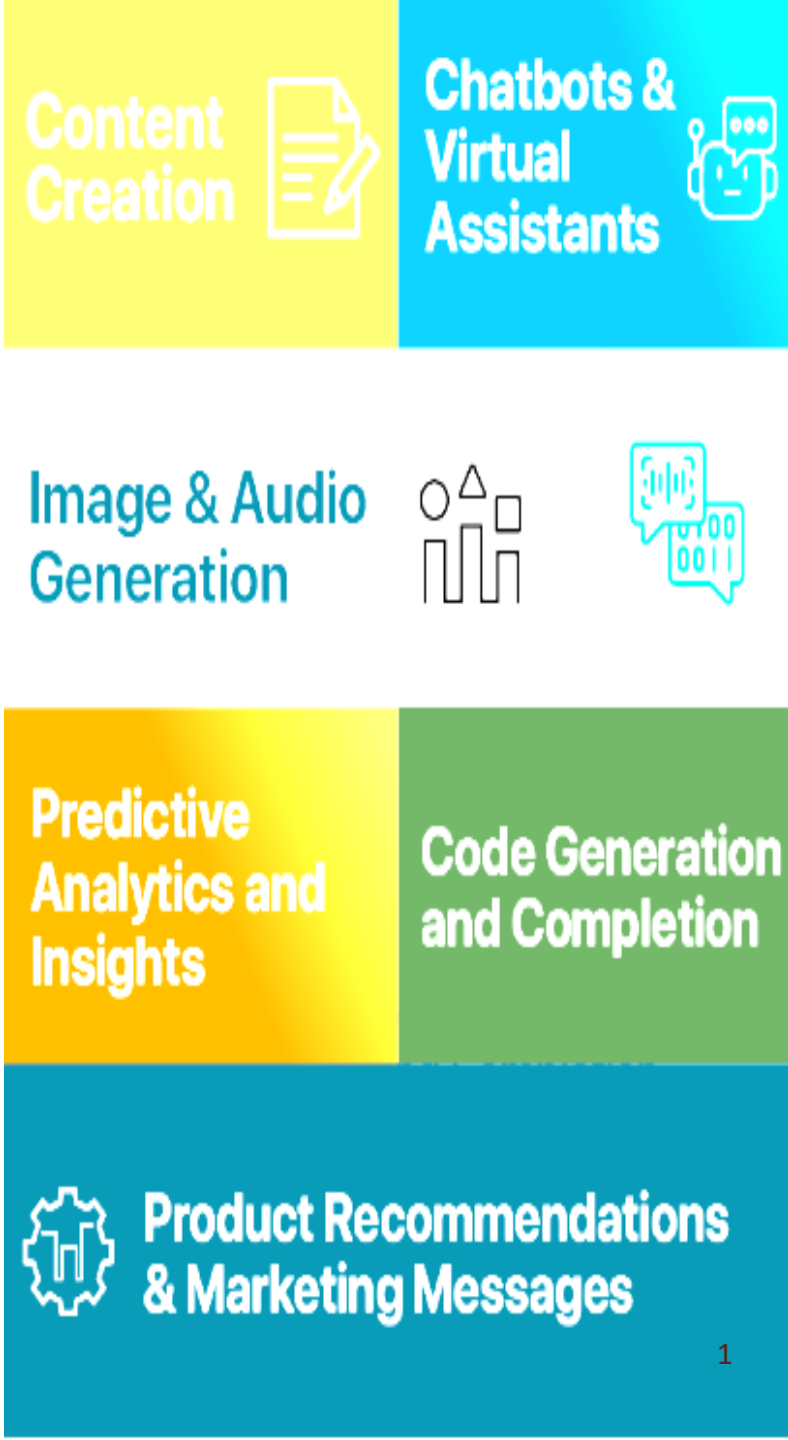
Gen-AI Models



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News Headlines…

- Gen-AI provides multiple opportunities for revenue generation

Resolve to use GenAI to make money



News Headlines…

FEATURES

In China, AI-Generated Fashion Models Are Hugely Popular — and Sexist



Sample images from Taji show how the AI tool can convert a product photo into a fashion image featuring a virtual model. From Alibaba

Learning Outcomes.

- Understand the theories behind Gen-AI and practically apply them with real-data sets

Outline

- Introduction to Gen-AI models
- Latent Variable Models
  - Autoencoders vs VAEs
  - GANs

Gen-AI



1



2

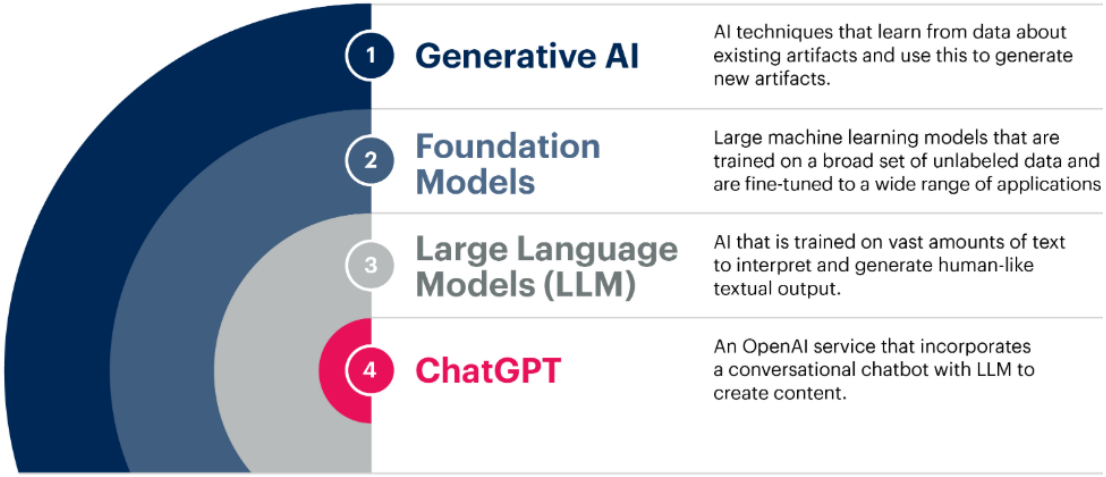


3

Gen-AI

Generative AI refers to unsupervised and semi-supervised machine learning algorithms that enable computers to use existing content like text, audio and video files, images, and even code to create new possible content.

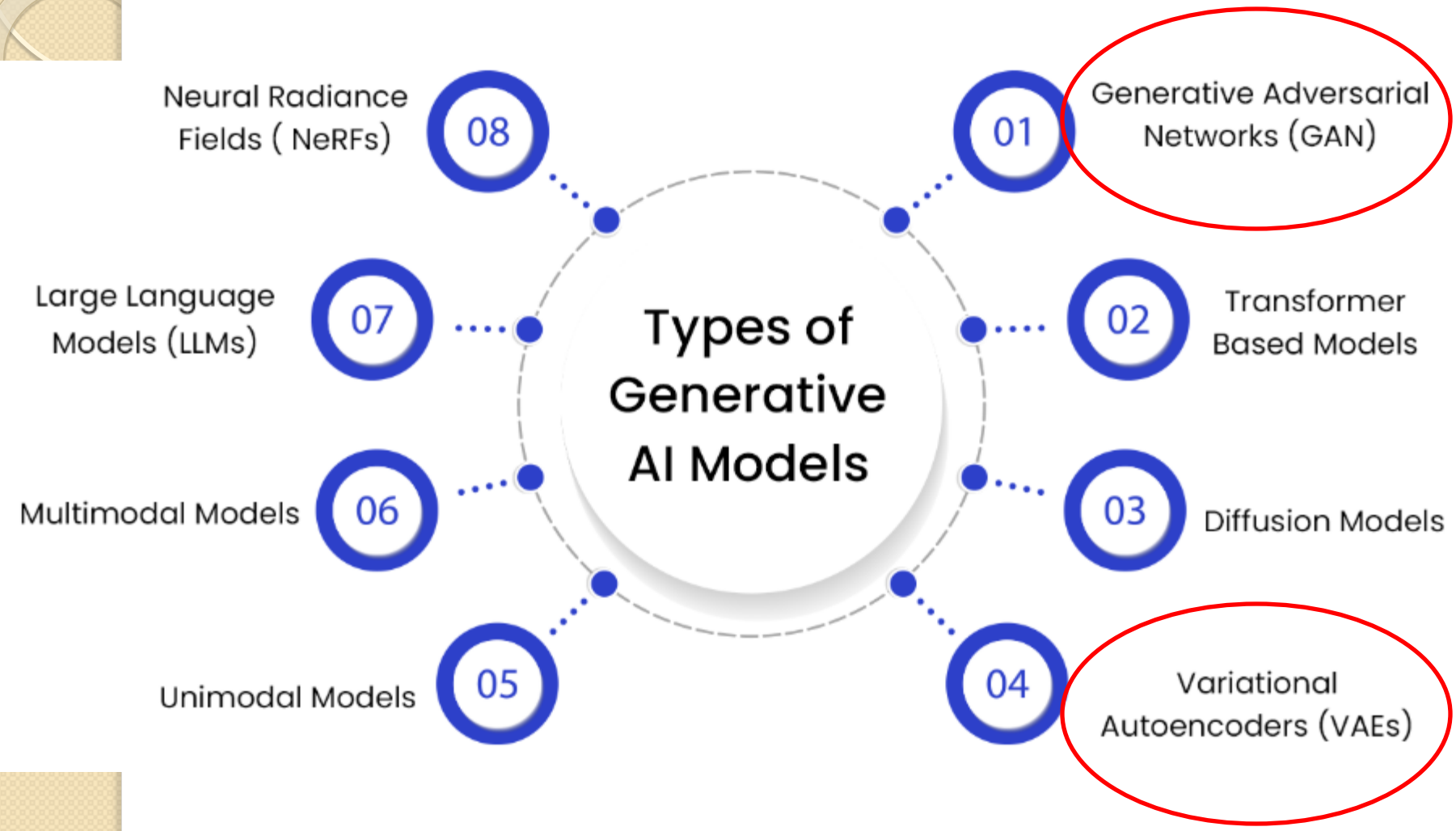
The main idea is to generate completely original artifacts that would look like the real deal.



Classification of ML models

- Supervised Learning
- Unsupervised Learning
- Semi-supervised Learning
- Reinforcement Learning

Gen-AI Models



Gen-AI Models

- What are latent variables?
  - **Hidden or unobserved variables that influence the observed data**.

    Eg:, if you want to generate realistic images of faces, you might use latent variables to capture the features, expressions, and styles of the faces.

  - Often represented by a low-dimensional vector, which can be seen as a compressed representation of the data.

Gen-AI Models

- Latent variable models
  - Autoencoders, Variational Autoencoders (VAEs), and β-VAE
  - Generative Adversarial Networks (GANs)

Unsupervised

Autoencoders..

- Autoencoders and Variational Autoencoders (VAEs)

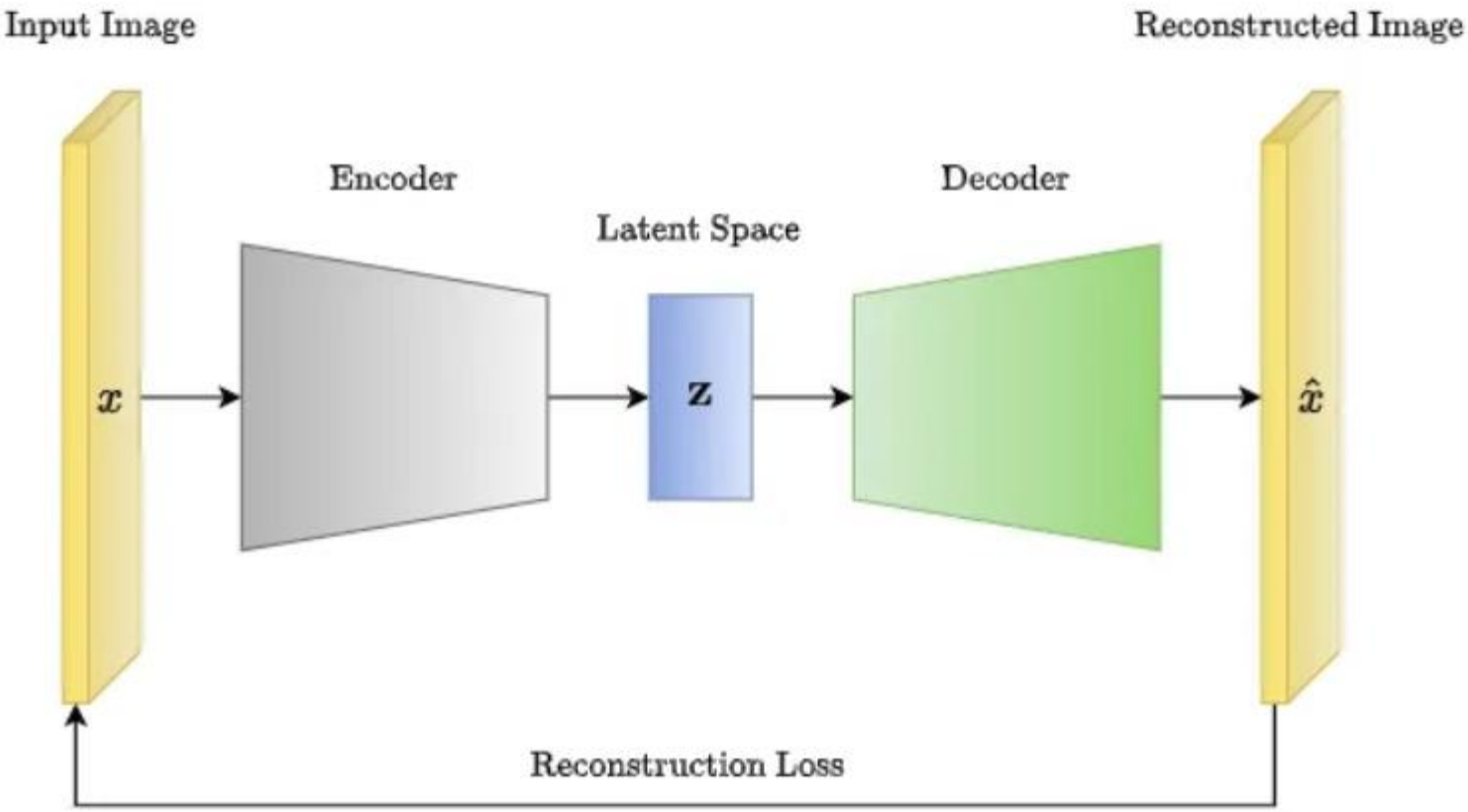
What is AE?

- Use to learn efficient embeddings (a NN designed to compress and then reconstruct data) of unlabeled data for a given network configuration.

- Objective: To minimize the difference between the input data and its reconstructed output, which is typically measured using a loss function such as mean squared error or binary cross-entropy

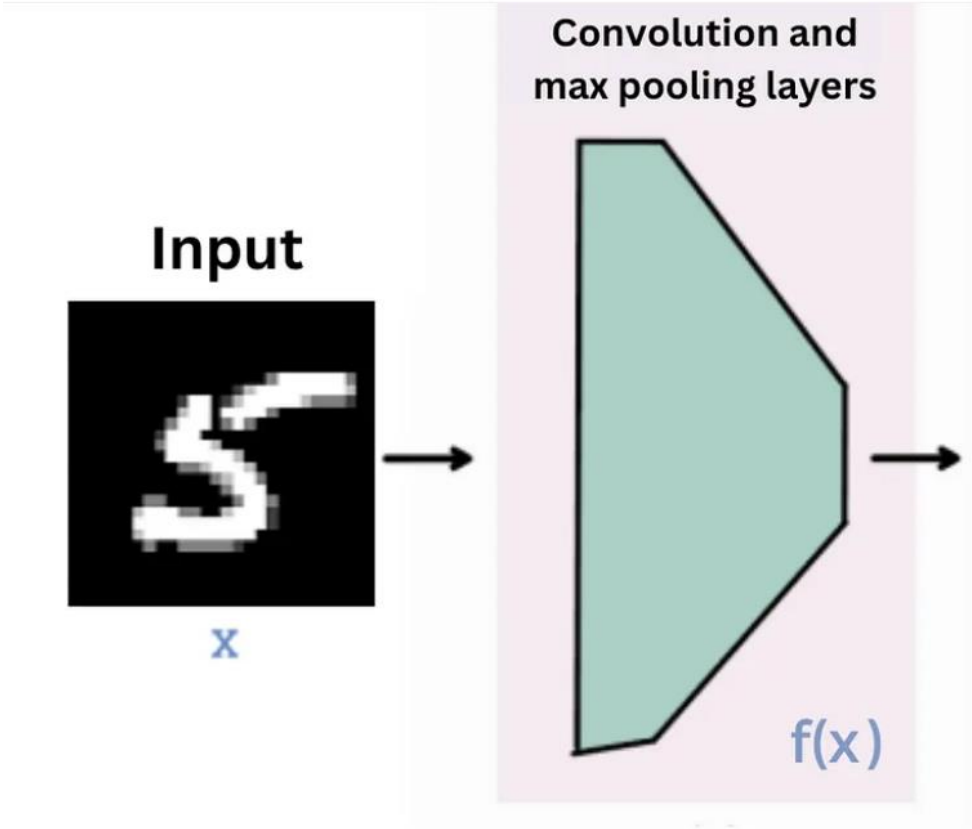
- Mainly used in applications with
  - Image denoising,
  - Dimensionality reduction,
  - Feature learning
  - Clustering
- Two components: (a)an encoder and (b)a decoder

Autoencoders…



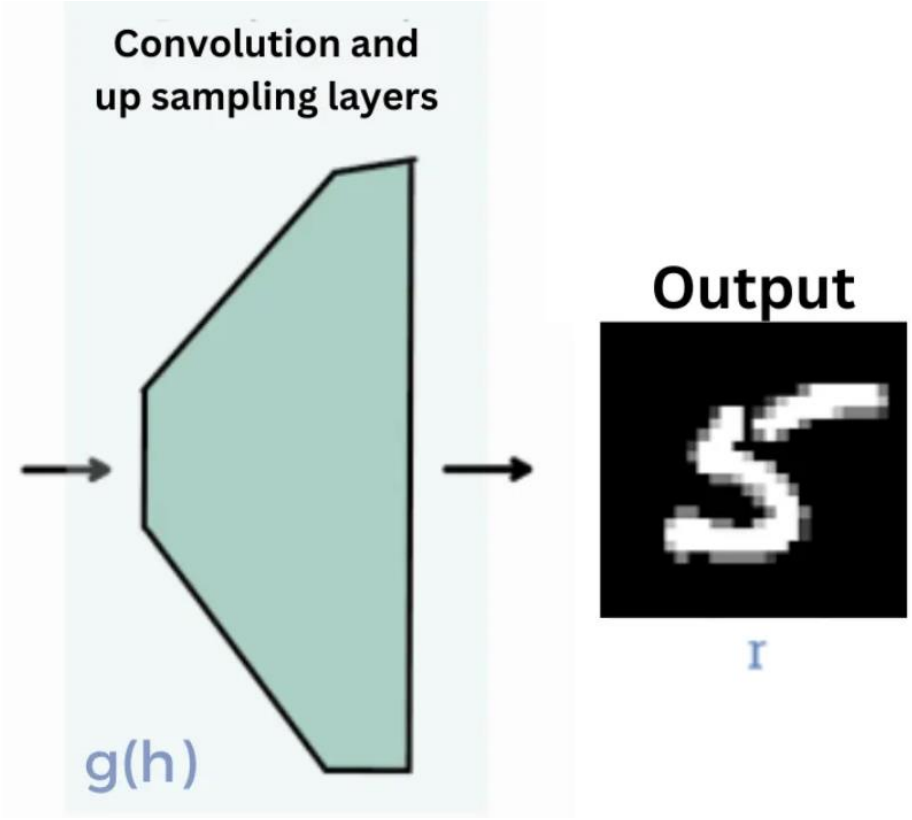
Autoencoders…

- Encoder
  - The encoder consists of a series of neural network layers (i.e. convolutional and pooling) which extract features from an image and embed or encode them to a low-dimensional latent space

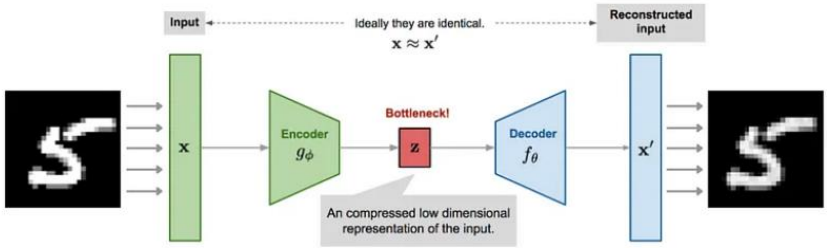


Autoencoders…

- Decoder
  - Acts as the "decompressor." It utilizes a sequence of up sampling and convolutional blocks to reconstruct the output based on the compressed knowledge representation stored in the bottleneck/latent vector



Autoencoders..



- Loss function
  - Mean Squared Error (MSE)

  A very common loss function used for training autoencoders.

  MSE is typically used **when the data is continuous**.

  This loss function calculates the square of the difference between the original input and the reconstructed output, and then takes the average of these squared differences.

$$Loss_{MSE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2$$

Autoencoders…

- Loss function
  - Binary Cross-Entropy (BCE)

  Also a common loss function especially used when **the output is binary or a probability distribution**.

  This loss function is very useful for classification tasks.

$$Loss_{BCE} = -\frac{1}{N}\sum_{n=1}^{N}\left[x_n \log \hat{x}_n + (1 - x_n)\log(1 - \hat{x}_n)\right]$$

Autoencoders- Limitations…

- Susceptible to overfitting (i.e. simply memorize the training data rather than learning to generalize to new data)
  - Adding regularization techniques such as dropout or early stopping

- Can be limited by the size of the compressed representation
  - Model needs a balance between **preserving the most relevant information** in the input and **minimizing the reconstruction error**.

Autoencoders…

- Exercise
  - Differentiate between reconstruction and regularization in latent variable models (i.e.autoencoders). Is there any trade-off?

Variational Autoencoders (VAEs)…

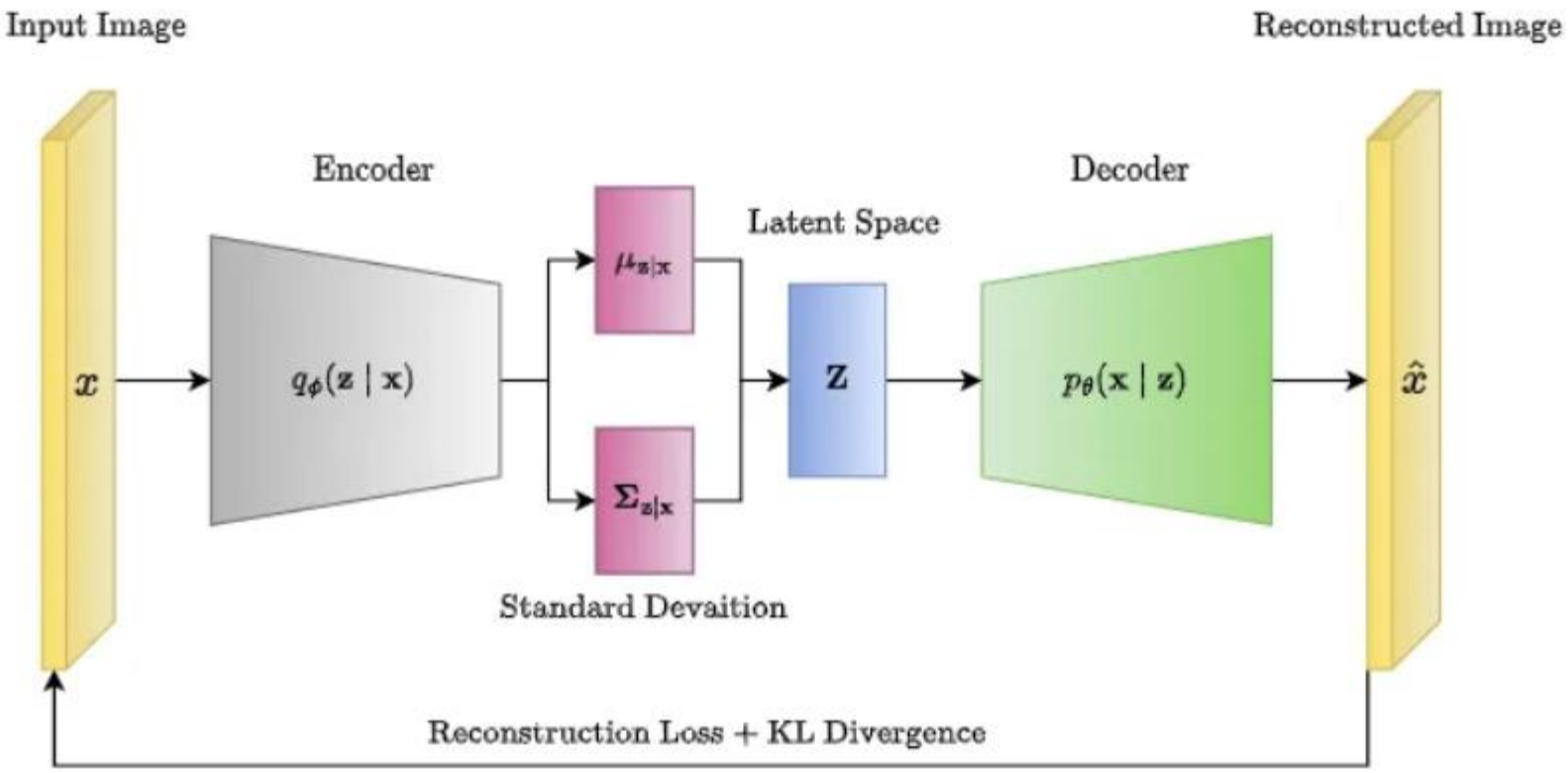
- VAEs addresses the issue of non-regularized latent space in autoencoder and provides the generative capability to the entire space

Variational Autoencoders (VAEs)…

- A type of autoencoder that was introduced to overcome some limitations of traditional AE

- VAEs extend the traditional AE architecture by introducing a probabilistic framework for generating the compressed representation of the input data

Variational Autoencoders (VAEs)…



Variational Autoencoders (VAEs)…

- Two key modifications required for AE
  1) Replace the encoder's output with a probability distribution

  Instead of the encoder outputting a point in the latent space, it **outputs the parameters of a probability distribution**, such as **mean and variance.** This distribution is typically a **multivariate Gaussian distribution** but can be some other distribution as well (e.g., Bernoulli).

Variational Autoencoders (VAEs)…

- Two key modifications required for AE
  1) Introduce a new term in the loss function called the Kullback-Leibler (KL) divergence

  This term measures the difference between the learned probability distribution over the latent space and a predefined prior distribution (usually a standard normal distribution).

  This term ensures that the learned distribution over the latent space is close to the prior distribution (This helps regularize the model and ensures that the latent space has a meaningful structure)

VAEs : Loss Function

- Composed of two parts: the reconstruction loss (similar to the traditional autoencoder loss) and the KL divergence loss

VAEs : Loss Function

- The reconstruction loss : measures the difference between the **original input** and the **output generated by the decoder**

- The KL divergence loss: measures the difference between the **learned probability distribution and the predefined prior distribution**

VAEs : Loss Function

$$\log p_\theta\left(x^{(i)}\right) \geq \mathcal{L}\left(x^{(i)}, \theta, \phi\right) = \underbrace{\mathbb{E}_z\left[\log p_\theta\left(x^{(i)} \mid z\right)\right]}_{\text{Reconstruct the Input Data}} - \underbrace{D_{KL}\left(q_\phi\left(z \mid x^{(i)}\right) \| p_\theta\left(z\right)\right)}_{\text{KL Divergence}}$$

$$\theta^*, \phi^* = \arg\max_{\theta,\phi} \sum_{i=1}^{N} \mathcal{L}\left(x^{(i)}, \theta, \phi\right)$$

The optimized term L in the above equation is called the ELBO (*E*xpectation *L*ower *BO*und).



- Assignment 01: 3 hours (Monday/ Tuesday)

Thank You